Repeated linear factors – 3.4

If we have a denominator like $\frac{p(x)}{x(x-2)}$,

we try to rewrite $\frac{p(x)}{x(x-2)} = \frac{A}{x} + \frac{B}{x-2}$.

If instead we had $\frac{p(x)}{x^2(x-2)}$, then this

cannot equal something of the above form.

Here we need to add a $\frac{C}{x^2}$ term.

If the denominator has a $(x-a)^n$ factor,

the partial fractions decomposition should include

terms of the form $\frac{A_1}{x-a} + \frac{A_2}{(x-a)^2} + \frac{A_3}{(x-a)^3} + \cdots + \frac{A_n}{(x-a)^n}$.

Example: Find $\int \frac{x-2}{(2x-1)^2(x-1)}\,dx$.

$$\frac{x-2}{(2x-1)^2(x-1)} = \frac{A}{2x-1} + \frac{B}{(2x-1)^2} + \frac{C}{x-1}$$

$$x-2 = A(2x-1)(x-1) + B(x-1) + C(2x-1)^2$$

$$x=1 \implies 1-2 = C \implies C=-1$$

$$x=\tfrac{1}{2} \implies \tfrac{1}{2}-2 = B\cdot -\tfrac{1}{2} \iff -\frac{3}{2} = -\frac{1}{2}B \iff B=3$$

$$x=0 \implies -2 = A - B + C = A-3-1 = A-4 \iff A=2.$$

$$\int \frac{x-2}{(2x-1)^2(x-1)}\,dx = 2\int \frac{1}{2x-1}\,dx + 3\int \frac{1}{(2x-1)^2}\,dx - \int \frac{1}{x-1}\,dx$$

$$u = 2x-1$$
$$du = 2\,dx$$

$$= 2 \cdot \frac{1}{2} \ln|2x-1| + 3 \cdot \frac{1}{2} \cdot \frac{1}{2x-1} - \ln|x-1| + C$$

$$= \ln|2x-1| + \frac{3}{4x-2} - \ln|x-1| + C$$

$$= \ln\left|\frac{2x-1}{x-1}\right| + \frac{3}{4x-2} + C$$